

EC1207

Principles of the Micro Economy

Winter 2023 Examination
Paper 1 · 1.5 hours · Section A: answer 2 of 3 (40%) · Section B: answer 1 of 2 (60%)

Module	EC1207 — Principles of the Micro Economy
Session	Winter 2023
Structure	Section A: three short questions, each with two 10% sub-parts — answer any two of the three. Section B: two long questions, each with three 20% sub-parts — answer any one of the two.
This scheme covers	All three Section A questions AND both Section B questions in full, so you can study every possible combination rather than just the ones you'd pick on the day.

How to use this scheme. Each full model answer is written to the depth an excellent script would reach — roughly 1.5 pages per 10% sub-part, with named real-world examples and academic or reputable citations throughout (Author, Year, Publication), matching the standard set out in the marking guidance for this module. The mark allocation table under each answer shows what separates a top-band script from a mid-band one, so you can see exactly where marks are actually earned, not just what the "correct" content is.

Section A — Short Questions

Answer 2 of 3 · 40% of total marks · each sub-part worth 10%

Question A1

20% Total

A1 (i)**10%**

Explain, using examples, why some goods do not follow the Law of Demand.

FULL MODEL ANSWER

The Law of Demand states that, *ceteris paribus*, as the price of a good rises, the quantity demanded falls — a negative price-quantity relationship represented by a downward-sloping demand curve. Three categories of good disobey this law, producing an upward-sloping demand curve, as shown in Figure 1.

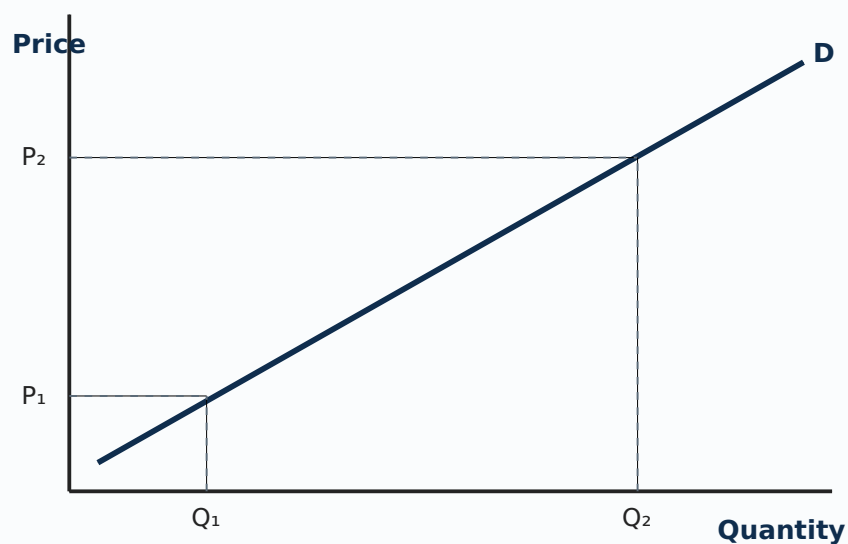


Figure 1: An upward-sloping demand curve. As price rises from P_1 to P_2 , quantity demanded RISES from Q_1 to Q_2 — the opposite of the Law of Demand.

1. Luxury / Snob Goods. These are goods perceived as high-status and exclusive. If the price rises, this reinforces the perception of exclusivity and demand actually increases; if the price falls, the good is no longer viewed as exclusive and demand falls. The high price is part of the appeal, not a deterrent. Classic examples include luxury cars, designer handbags, and luxury yachts. This concept was first theorised by Thorstein Veblen (1899, *The Theory of the Leisure Class*), who coined the term “conspicuous consumption” to describe purchases made primarily to signal wealth and status to others — which is exactly why a price cut can backfire: if Louis Vuitton significantly cut prices, demand could fall by destroying the exclusivity signal that justifies the purchase in the first place.

2. Goods Based on Expected Values. The most prominent examples come from financial markets — stocks, and more recently Bitcoin. If the price of a share rises, this can signal to investors that the asset is a good investment, or that further gains are likely, so they demand *more* of it rather than less; if the price falls, investors may demand less, expecting the decline to continue. Demand here is driven by expectations of further price movement, not by the price's effect on affordability — the reverse of how the Law of Demand normally operates. Ireland's own housing market during the 2000s boom illustrated a closely related phenomenon: rising prices intensified buyer demand as purchasers expected continued appreciation, rather than being priced out (Shiller, 2015, *Irrational Exuberance*).

3. Giffen Goods. A Giffen good is an inferior good for which demand actually rises as price rises, because the negative income effect dominates and overrides the substitution effect. Consider a low-income household whose basic foodstuff is bread: if the price of bread increases, the household becomes poorer in real terms and can no longer afford the more expensive alternative (meat) — so they are forced to buy *more* bread, since it remains the only food they can afford, even though its price has risen. This was famously illustrated by potato consumption during the Irish Famine, and confirmed empirically in a modern setting for rice consumption in Hunan province, China (Jensen & Miller, 2008, *American Economic Review*). Importantly, all Giffen goods are inferior goods, but not all inferior goods are Giffen goods — this depends specifically on how demand reacts to a change in *price*, not just to a change in income.

How the 10% Is Likely Allocated

Tier	Marks	What separates this tier
9-10%	9-10%	Two or more exceptions identified, each with the economic mechanism clearly explained, a specific real-world example, and (ideally) an academic or reputable citation. Diagram (or clear description of an upward-sloping demand curve) included.
6-8%	6-8%	At least one exception well explained with mechanism and example; a second exception present but less developed, or citations weak/missing.
3-5%	3-5%	Exceptions named but the underlying mechanism not properly explained; examples vague or generic.
0-2%	0-2%	No meaningful attempt, or fundamental confusion about what an exception to the Law of Demand actually means.

A1 (ii)**10%**

Outline, using illustrations, two factors that may cause the demand curve to shift to either the right or the left.

FULL MODEL ANSWER

A shift of the whole demand curve (as opposed to a movement *along* it, which is only caused by a change in the good's own price) is caused by a change in a non-price determinant of demand. Two of the most important are income and the price of substitute goods.

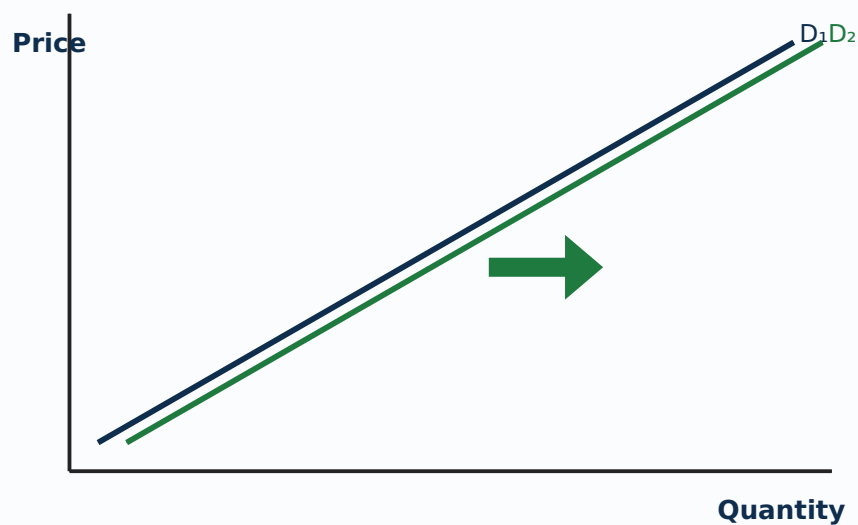


Figure 2: A rightward shift of the demand curve ($D_1 \rightarrow D_2$) — at every price, quantity demanded is now higher.

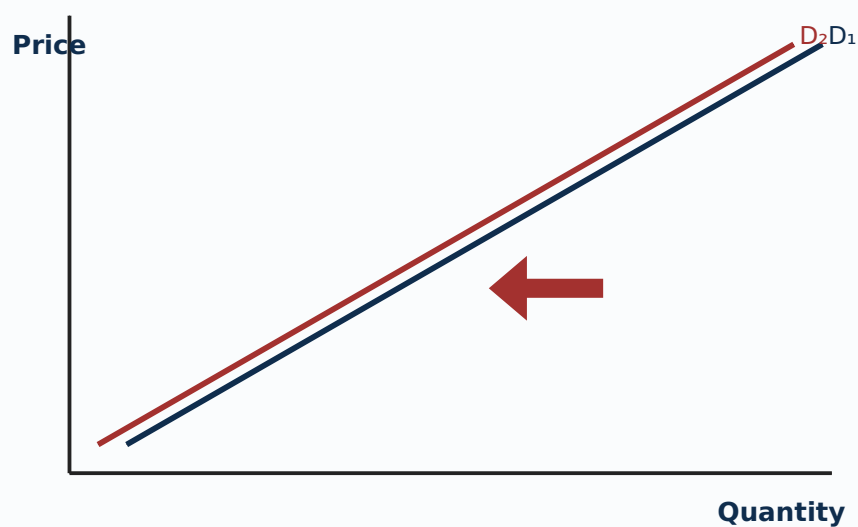


Figure 3: A leftward shift of the demand curve ($D_1 \rightarrow D_2$) — at every price, quantity demanded is now lower.

1. Income

For a normal good. A rise in income increases consumers' purchasing power, causing the demand curve to shift **rightward** (Figure 2) — consumers now demand more of the good at every price. A fall in income shifts demand **leftward** (Figure 3). Most goods and services behave this way — for example, demand for restaurant meals or new clothing typically rises as household income rises.

For an inferior good, the relationship reverses. When income rises, demand for an inferior good actually falls (a negative income effect) — the demand curve shifts **leftward** even though consumers are better off. Examples include own-brand supermarket goods, frozen ready meals, and bus travel: as income rises, consumers often switch toward branded alternatives, restaurant meals, or car ownership instead, reducing demand for the inferior alternative.

2. Price of a Substitute Good

Substitutes are goods that are interchangeable or replaceable in consumption. Examples include Sprite and 7up, tea and coffee, or McDonald's and Burger King. An **increase** in the price of one substitute causes consumers to switch toward the relatively cheaper alternative, shifting the **other** good's demand curve **rightward** (Figure 2) — if the price of tea rises, demand for coffee shifts right as consumers switch. Conversely, a **decrease** in the price of one substitute shifts demand for the other good **leftward** (Figure 3), as consumers are drawn toward the now cheaper alternative instead.

Other valid factors include the price of a complement, a change in preference/taste, and a change in the number of consumers in the market — any two of these five, correctly illustrated, are creditable.

How the 10% Is Likely Allocated

Tier	Marks	What separates this tier
9-10%	9-10%	Two genuinely distinct factors identified, each correctly illustrated with a labelled shift diagram (rightward and/or leftward as appropriate) and a specific example, with the distinction between normal and inferior goods correctly drawn out for the income factor if chosen.
6-8%	6-8%	Two factors identified and explained correctly, but with only one diagram, or diagrams present but not clearly labelled.
3-5%	3-5%	One factor well explained; the second vague or confused with a movement along the curve rather than a shift.
0-2%	0-2%	No meaningful attempt, or confusion between a shift of the curve and a movement along it.

Question A1 Total: 20%

Question A2

20% Total

A2 (i)

10%

Describe, using an example, how the sunk cost fallacy can affect consumers' decisions.

FULL MODEL ANSWER

A **sunk cost** is a cost that has already been incurred and cannot be recovered, regardless of any future decision. Standard economic theory (Mankiw, 2006) is unambiguous that sunk costs should be irrelevant to forward-looking decisions — only future costs and benefits should determine behaviour, since the money already spent is gone either way.

The Sunk Cost Fallacy. The sunk cost fallacy occurs when individuals allow a past, unrecoverable expenditure to irrationally influence a future decision, rather than evaluating only the costs and benefits that still lie ahead. This is one of the most robust findings in behavioural economics, and is driven primarily by **loss aversion**: abandoning something already paid for feels like crystallising a loss, which is psychologically uncomfortable, while continuing — even when economically irrational — provides the illusion of not having wasted the money.

Consumer Example — The Westlife Concert Ticket. Imagine paying €60 for a non-refundable ticket to see Westlife in Croke Park. On the day of the concert, heavy snow makes the drive to Dublin genuinely dangerous. The rational, forward-looking decision weighs only the risk of the journey against the enjoyment of attending — the €60 is spent regardless of whether the concert is attended or not, so it should play no role in the decision at all. The sunk cost fallacy nonetheless pushes many people to make the risky journey specifically *because* they already paid, when they would very likely decide not to travel at all if the ticket had been given to them for free.

Business Application — The Concorde Effect. The fallacy extends beyond individual consumers. The Concorde supersonic airliner project was already predicted to be commercially unviable early in its development, yet successive British and French governments continued funding it for decades, unable to abandon a project into which so much taxpayer money had already been sunk — they finally gave up only in 2003, having spent a total of £1.3 billion. This became so well documented that it is now known in the academic literature as the “Concorde Fallacy” (Dawkins & Carlisle, 1976, *Nature*).

A complete answer explicitly names loss aversion as the psychological driver, rather than simply describing the fallacy's symptoms — the mechanism (why the past cost *feels* relevant even though it isn't) is what separates a strong answer from one that merely restates the definition.

How the 10% Is Likely Allocated

Tier	Marks	What separates this tier
9-10%	9-10%	Clear definition of sunk cost and the fallacy, the psychological mechanism (loss aversion) explicitly explained, at least one fully worked example with the mechanism applied directly to the consumer's decision, and an academic or reputable citation.
6-8%	6-8%	Correct definition and one good example, but the psychological underpinning (loss aversion) not clearly explained, or no citation.
3-5%	3-5%	Sunk cost concept partially understood; example present but not clearly linked to the mechanism of the fallacy.
0-2%	0-2%	No meaningful attempt, or the concept confused with opportunity cost.

A2 (ii)**10%**

Explain, using an example, how anchoring can affect consumers' decisions.

FULL MODEL ANSWER

Anchoring is a cognitive bias that leads individuals to rely too heavily on the first piece of information they receive when making a decision or estimate, even when that piece of information is arbitrary or irrelevant to the true value of what's being judged.

The Classic Demonstration — Tversky and Kahneman's Wheel of Fortune (1974). In one of the founding experiments on anchoring, participants spun a wheel of fortune rigged to land only on 10 or 65, and were then asked whether the percentage of African countries in the United Nations was higher or lower than that number, before estimating the actual percentage. Participants who landed on 10 guessed an average of around 25%; those who landed on 65 guessed around 45% — despite the wheel result being entirely random and having no genuine bearing on the correct answer, it powerfully anchored participants' subsequent estimates.

Consumer Example — Buying a Beer on the Beach. Imagine lying on a beach on a hot day; a friend offers to fetch a beer and asks how much you'd be willing to pay, telling you it will come from either a fancy hotel or a small run-down off-licence. The location named — the only piece of information you have — anchors your expectation of a reasonable price before you've considered anything else about the beer itself. Naming the fancy hotel anchors consumers toward a much higher acceptable price than naming the off-licence would, even though the actual product (a beer) may be identical.

Retail Pricing — The JC Penney Example. When JC Penney CEO Ron Johnson eliminated coupons and marked-down reference prices in favour of “everyday low pricing,” sales collapsed — removing the anchor (the higher original price shoppers used to judge whether they were getting a bargain) removed the very thing that made a lower price feel rational and rewarding. Johnson was fired within 17 months of the policy's introduction, and the company reversed course, reintroducing anchor prices and coupons. This illustrates that consumers don't just want a low price — they want to see the anchor that tells them the price they're paying is a genuine deal.

Anchoring is closely related to, but distinct from, **framing** (how information is presented, rather than which initial number is provided) — a strong answer keeps the two concepts separate, since conflating them is a common error.

How the 10% Is Likely Allocated

Tier	Marks	What separates this tier
9-10%	9-10%	Accurate definition of anchoring, at least one fully worked consumer example with the mechanism clearly explained, and reference to the Tversky and Kahneman (1974) experimental foundation or an equivalent citation.
6-8%	6-8%	Correct definition and one good example with the mechanism explained, but no reference to the underlying experimental research.
3-5%	3-5%	Anchoring loosely defined; example present but the mechanism not clearly linked to the consumer's decision.
0-2%	0-2%	No meaningful attempt, or anchoring confused with framing or another bias.

Question A2 Total: 20%

Question A3

20% Total

Suppose the price of orange juice increases from €1.10 to €1.65 and the quantity demanded of apple juice increases from 45 litres to 65 litres.

A3 (i)**10%**

(i) Calculate the cross-price elasticity of demand for apple juice in response to a change in the price of orange juice.

FULL MODEL ANSWER

Cross-Price Elasticity of Demand (XED) = % Change in Quantity Demanded of Good B ÷ % Change in Price of Good A

Step 1 — % Change in Quantity Demanded of Apple Juice

$$\% \Delta Q_d = (65 - 45) \div 45 \times 100 = 20 \div 45 \times 100 = +44.44\%$$

Step 2 — % Change in Price of Orange Juice

$$\% \Delta P = (1.65 - 1.10) \div 1.10 \times 100 = 0.55 \div 1.10 \times 100 = +50.00\%$$

Step 3 — XED

$$\text{XED} = +44.44\% \div +50.00\% \approx +0.89$$

The cross-price elasticity of demand for apple juice with respect to the price of orange juice is approximately **+0.89**.

How the 10% Is Likely Allocated

Tier	Marks	What separates this tier
9-10%	9-10%	Both percentage changes calculated correctly and shown step by step. XED formula stated clearly before substitution. Final value correct ($\approx +0.89$). All working legible.
6-8%	6-8%	Formula correct and method sound but a minor arithmetic error. Working clearly shown.
3-5%	3-5%	Some understanding of the XED formula but significant computational errors. Partial working present.
0-2%	0-2%	No meaningful attempt, or a completely incorrect formula applied.

A3 (ii)**10%**

(ii) What does the cross-price elasticity of these goods, estimated in part (i) above, tell you about the relationship between these two goods?

FULL MODEL ANSWER

Interpreting the sign — substitutes. The positive sign of $XED \approx +0.89$ is the most immediately important result. A positive XED indicates that orange juice and apple juice are **substitute** goods — goods that can be used in place of one another. When the price of orange juice rises, some consumers switch to the relatively cheaper alternative (apple juice), increasing its quantity demanded. This is the defining characteristic of substitutes: a rise in the price of one good increases demand for the other.

Interpreting the magnitude. The magnitude of $+0.89$ indicates a fairly strong, though not quite “elastic,” substitution relationship. Since $|XED|$ is close to but just under 1, a 1% rise in the price of orange juice generates just under a 1% rise in the quantity demanded of apple juice — a roughly proportionate response. This makes intuitive sense: both are fruit juices, commonly stocked side by side in the same section of a supermarket, and a consumer who simply wants a fruit juice with breakfast has little reason to insist on orange juice specifically once its price rises noticeably relative to apple juice.

Cross-Price Elasticity Classification

- $XED > 0$ (Positive) → Substitute goods — e.g. orange juice and apple juice, tea and coffee, Sprite and 7up
- $XED < 0$ (Negative) → Complementary goods — e.g. cars and petrol, tennis rackets and tennis balls, printers and printer ink
- $XED = 0$ → Unrelated / independent goods — e.g. orange juice and car tyres

Business strategy implications. For an orange juice producer, this result has real commercial significance: any price increase must be handled cautiously, since a 1% rise will push demand toward apple juice by almost the same proportion. This reinforces the importance of non-price competition — brand differentiation, loyalty schemes, and product innovation (fresh, no-pulp, vitamin-fortified variants) — to reduce the substitution elasticity and retain consumers even as prices rise (Kotler & Keller, 2016, *Marketing Management*).

How the 10% Is Likely Allocated

Tier	Marks	What separates this tier
9-10%	9-10%	Positive sign correctly interpreted as substitutes with the mechanism explained. Magnitude discussed appropriately. All three XED cases referenced (substitute/complement/independent). Business implications discussed. Citation present.
6-8%	6-8%	Correct identification of substitutes from the positive XED. Magnitude noted. Limited business application or no citation.
3-5%	3-5%	Identifies substitutes but the explanation of the mechanism is incomplete or partially incorrect.
0-2%	0-2%	No meaningful attempt, or the wrong relationship identified from the XED value.

Question A3 Total: 20%

Section B — Long Questions

Answer 1 of 2 · 60% of total marks · each sub-part worth 20%

Question B1

60% Total

B1 (i)

20%

(i) Using the data in Table 1 below, determine the optimum level of output for a firm in perfect competition. Explain your answer.

Table 1: Total Revenue and Total Cost for Firm A, by Output Level

Quantity Produced	Total Revenue (€)	Total Cost (€)
4	170	141
5	200	160
6	230	190
7	260	211
8	290	247

FULL MODEL ANSWER

A firm in perfect competition maximises profit at the output level where the gap between total revenue and total cost is largest — equivalently, where marginal revenue (MR) still exceeds or just equals marginal cost (MC), and producing one more unit would not add further profit.

Step 1 — Profit at Each Output Level

Quantity	TR (€)	TC (€)	Profit (€)
4	170	141	29
5	200	160	40
6	230	190	40
7	260	211	49
8	290	247	43

Step 2 — Cross-Check via Marginal Revenue and Marginal Cost

Moving from Q to Q+1	$\Delta TR = MR$ (€)	$\Delta TC = MC$ (€)	Worth producing?
4 → 5	30	19	Yes — $MR > MC$, adds €11 profit
5 → 6	30	30	Indifferent — $MR = MC$, adds €0
6 → 7	30	21	Yes — $MR > MC$, adds €9 profit
7 → 8	30	36	No — $MC > MR$, reduces profit by €6

Total revenue rises by exactly €30 for every extra unit sold — consistent with perfect competition, where the firm is a **price taker** facing a constant price ($P = AR = MR = €30$). Marginal cost, by contrast, varies unit to unit: €19, €30, €21, then €36.

Producing the 7th unit is worthwhile because its marginal cost (€21) is below the €30 marginal revenue it generates, adding €9 to profit (taking total profit from €40 at $Q=6$ to €49 at $Q=7$). Producing the 8th unit is **not** worthwhile, since its marginal cost (€36) exceeds the €30 marginal revenue it would generate, which would reduce profit by €6 (from €49 back down to €43).

Conclusion

The optimum (profit-maximising) level of output is $Q = 7$ units, generating a maximum profit of €49.

Both methods agree exactly, which is itself a useful check: the total-profit comparison (Step 1) directly identifies €49 at $Q=7$ as the highest profit figure in the table, while the MR/MC marginal comparison (Step 2) confirms this is precisely the last unit for which MR still exceeds MC — the textbook profit-maximising rule of “produce up to the point where $MR = MC$, and no further.”

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
17-20%	17-20%	Profit correctly calculated at every output level, $Q=7$ correctly identified as the profit-maximising output with €49 the maximum profit, AND the answer is cross-checked / explained via the $MR=MC$ marginal rule, not simply read off the table.
12-16%	12-16%	$Q=7$ correctly identified as optimal from the profit column, but the MR/MC marginal reasoning is not used to explain why.
7-11%	7-11%	Profit calculated for most output levels but the wrong output level selected as optimal (e.g. choosing $Q=6$, where profit ties with $Q=5$ rather than checking $Q=7$).
3-6%	3-6%	Table partially completed with some arithmetic errors; method unclear.
0-2%	0-2%	No meaningful attempt.

B1 (ii)**20%**

(ii) Compare and contrast perfectly competitive markets and oligopoly markets in terms of their potential costs and benefits to society. Use examples to illustrate your answer.

FULL MODEL ANSWER**Perfectly Competitive Markets**

Structure and long-run equilibrium. Perfect competition assumes homogeneous products, many buyers and sellers, free entry and exit, and perfect information — conditions that make every firm a **price taker** facing a horizontal ($P = AR = MR$) demand curve. In long-run equilibrium, free entry and exit compete away any supernormal profit, so $P = MC = \text{minimum ATC}$: the firm earns only normal profit.

Benefits to society. This delivers **allocative efficiency** ($P = MC$, so resources flow to their highest-valued use with no deadweight loss) and **productive efficiency** (output produced at the lowest possible unit cost). Consumer surplus is maximised, and there is no sustained exploitation of consumers through supernormal profit. Agricultural commodity markets, such as many fresh produce markets, approximate this model.

Costs and limitations. Homogeneous products mean no variety for consumers with differentiated preferences, and because supernormal profit is competed away almost immediately, firms have little financial incentive to innovate — perfect competition may be statically efficient but dynamically weak.

Oligopoly Markets

Structure. An oligopoly is dominated by a small number of large firms, offering either differentiated (e.g. Coca-Cola and Pepsi) or largely homogeneous (e.g. petrol) products. The defining feature is **interdependence**: each firm's pricing and output decisions must account for how rivals are likely to react — a strategic dimension entirely absent in perfect competition. This is often modelled with a **kinked demand curve** (Figure 4): if a firm raises its price, rivals don't follow and it loses substantial market share (demand is elastic above the kink); if it cuts price, rivals match the cut to avoid losing customers, so quantity demanded rises only modestly (demand is inelastic below the kink). This tends to make the prevailing price P^* “sticky” — oligopolists have little incentive to deviate from it in either direction.

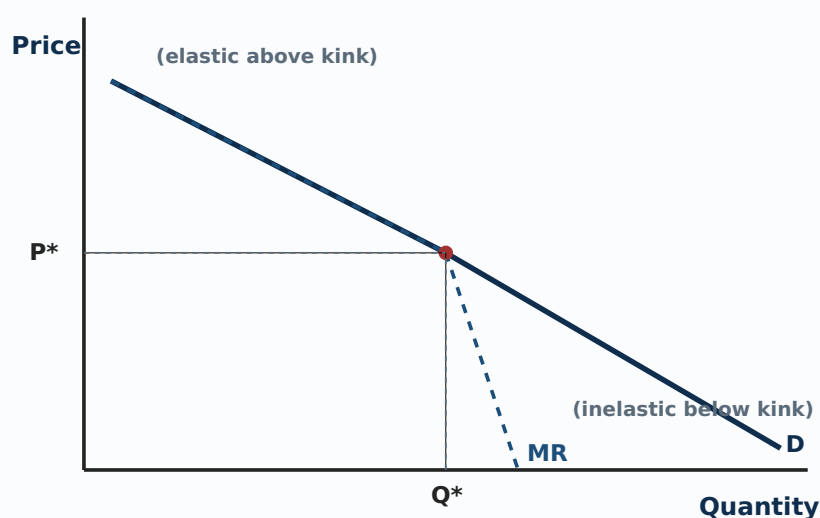


Figure 4: The kinked demand curve facing an oligopolist. Above the kink (price rises), rivals do not follow, so demand is relatively elastic; below the kink (price cuts), rivals match the cut, so demand is relatively inelastic — the price P^ at the kink tends to be sticky.*

Benefits to society. Product differentiation gives consumers genuine variety, and the profit incentive under oligopoly (unlike under perfect competition) can support meaningful investment in innovation and product improvement. Where firms compete vigorously on non-price dimensions — product quality, service, advertising — this can generate real consumer value.

Costs to society. Because oligopolists set price above marginal cost and output below the competitive level, oligopoly is **allocatively inefficient**. Firms may also collude, explicitly or tacitly, forming a **cartel** that behaves like a monopoly — restricting output and raising prices well above the competitive level, directly harming consumers. The classic **Prisoners' Dilemma** illustrates the tension: each firm's dominant strategy is to undercut the other (or advertise, or expand output), even though both would be better off cooperating — competition law exists specifically to prohibit the explicit collusion that would otherwise let oligopolists reach the more profitable, but socially worse, monopoly outcome.

Critical Comparison

Which serves society better depends on what's being weighed. On static efficiency grounds, perfect competition is unambiguously superior: $P = MC$ and zero deadweight loss are simply not features of oligopoly, where P consistently exceeds MC . But perfect competition's lack of any innovation incentive is a genuine, permanent weakness — and in the real world, homogeneous products and perfect information are rare, meaning perfect competition is best understood as a theoretical benchmark rather than a market structure society could realistically adopt across most industries. Oligopoly, despite its inefficiencies, is the structure that actually characterises most real consumer-facing industries (supermarkets, telecoms, airlines), and the innovation and variety it can support may partially offset its static inefficiencies — though this depends heavily on whether firms genuinely compete or manage, through tacit collusion, to behave as a de facto monopoly.

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
17-20%	17-20%	Both market structures explained accurately (including the kinked demand curve for oligopoly, correctly diagrammed), with both benefits AND costs to society articulated for each, specific examples used throughout, and a critical comparison reaching a reasoned view on which better serves society and under what conditions.
12-16%	12-16%	Both structures compared across several dimensions with reasonable examples, but the critical comparison is thinner or the diagram is missing/unclear.
7-11%	7-11%	Structures described largely in isolation rather than directly contrasted; limited evaluation.
3-6%	3-6%	Surface-level treatment with significant errors or confusion between the two structures.
0-2%	0-2%	No meaningful attempt.

B1 (iii)**20%**

(iii) Using examples, discuss the barriers of entry to monopoly markets and explain how they differ from other market types.

FULL MODEL ANSWER

A monopoly exists where a single seller faces no close substitutes and is protected by **barriers to entry** — structural or legal obstacles that prevent rival firms from entering the market and competing away the monopolist's supernormal profit. Three main types are recognised.

1. Monopoly Resources

A key input or resource is owned by a single firm or body. Where one entity controls a resource essential to producing the good, no rival can enter regardless of how profitable the market looks. In Ireland, the State owns the national rail network (Iarnród Éireann effectively holds a monopoly resource in the rail lines themselves), and local county councils control the public water supply infrastructure — a private firm cannot simply build a competing rail line or water network alongside the existing one.

2. Government-Created Monopolies

The government grants an exclusive legal right to produce a good or service. Patents give the holder exclusive rights to manufacture and sell an invention (such as a new drug) for a fixed number of years, and copyright grants creators exclusive control over how an original work is used. A more direct example is Sweden's state-owned retail alcohol monopoly, Systembolaget, which the government deliberately created and protects from competition specifically to reduce alcohol-related illness, fatal accidents, and suicides — research suggests privatising it would increase availability, advertising, and overall alcohol demand, illustrating that this barrier to entry can be a deliberate public health tool, not merely an economic accident.

3. Low Costs of Production (Natural Monopoly)

A single large producer can supply the entire market more cheaply than several smaller competing firms. Where a first mover achieves very low unit costs (often through large fixed infrastructure costs spread over a very large output), high start-up costs deter any rival from entering profitably, even without government protection. This barrier can also be reinforced through mergers, acquisitions, or collusion that consolidate the industry into a single dominant producer over time.

How Monopoly Differs From Other Market Types

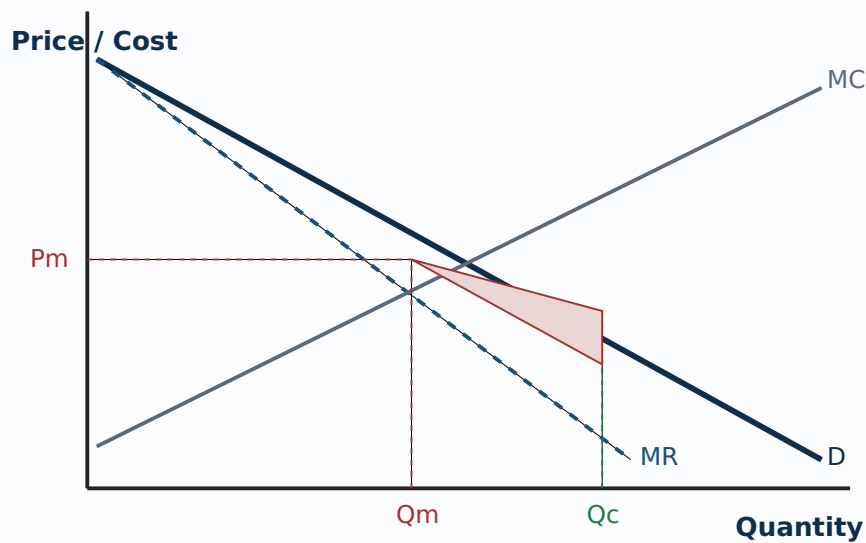


Figure 5: The monopoly welfare loss. Setting $MR = MC$ gives monopoly output Q_m at price P_m ; the competitive outcome ($P = MC$) would occur at the higher output Q_c . The shaded triangle is the deadweight loss from monopoly restriction of output.

Output and pricing behaviour. Like a firm in perfect competition or an oligopolist, a monopolist maximises profit where $MR = MC$ — but because the monopolist's demand curve *is* the entire market demand curve (there being no other seller), price is set above marginal cost ($P > MR = MC$), unlike perfect competition where $P = MR = MC$ exactly. This produces the deadweight welfare loss shown in Figure 5: monopoly output (Q_m) falls short of the competitive output (Q_c), and the shaded triangle represents transactions between willing buyers and sellers that are blocked purely by the monopolist's output restriction.

Entry barriers themselves are the key structural difference. Perfect competition assumes free entry and exit, guaranteeing only normal profit in the long run. Oligopoly has significant, but not absolute, barriers to entry (economies of scale, patents, brand proliferation) — a small number of rivals can and do compete, and interdependence (strategic reaction to rivals) is the defining feature. Monopoly's barriers are absolute enough that no rival enters at all, which is precisely why a monopolist, unlike a perfectly competitive firm, can sustain supernormal profit even in the long run.

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
17-20%	17-20%	All three barrier types explained with specific named examples (Irish rail/water, Systembolaget or patents/copyright, natural monopoly), the monopoly pricing outcome ($P > MR = MC$) correctly diagrammed with deadweight loss shown, and a clear comparison of how entry barriers differ in degree across perfect competition, oligopoly, and monopoly.
12-16%	12-16%	Barrier types covered with examples but the comparison to other market types is thinner, or the diagram is missing/unclear.
7-11%	7-11%	One or two barrier types described with limited examples; comparison to other market structures largely absent.
3-6%	3-6%	Surface-level treatment with significant gaps or errors.
0-2%	0-2%	No meaningful attempt.

Question B1 Total: 60%

Question B2

60% Total

B2 (i)**20%**

(i) Discuss, using examples, one positive and one negative externality of business operations in Ireland. Your examples may be from any industry. Use appropriate diagrams to illustrate your answer.

FULL MODEL ANSWER**Negative Externality — Agricultural Greenhouse Gas and Ammonia Emissions**

Ireland's agriculture sector, and dairy farming in particular, produces a substantial negative externality through greenhouse gas and ammonia emissions.

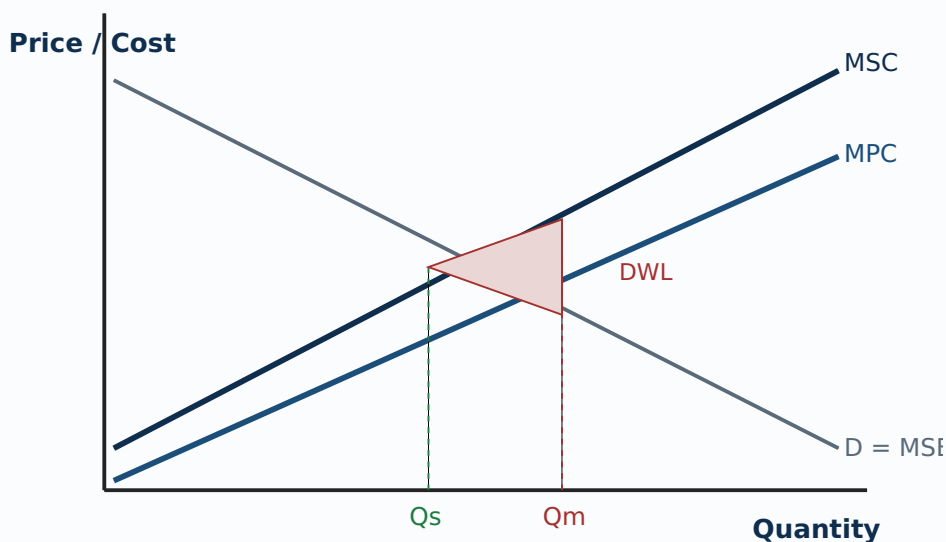


Figure 6: Negative production externality. The free market produces at Q_m (where $MPC = MSB$), overproducing relative to the social optimum Q_s (where $MSC = MSB$). The shaded triangle is the deadweight welfare loss from overproduction.

The mechanism. When a dairy farm considers how much milk to produce, it weighs only its own Marginal Private Cost (MPC) — feed, labour, land, veterinary care. But dairy farming also generates methane from enteric fermentation in cattle, nitrous oxide from fertiliser and manure, and ammonia emissions, none of which the farm pays for directly. These costs are borne by wider society (through climate damage and air quality impacts) but are absent from the farm's own cost calculation, so the Marginal Social Cost (MSC) of milk production exceeds its MPC. As Figure 6 shows, the free market produces at Q_m (where $MPC = MSB$), but the socially optimal output is the lower Q_s (where $MSC = MSB$) — the market **overproduces** relative to the social optimum, generating a deadweight welfare loss.

The scale of the problem in Ireland. Agriculture is Ireland's single largest greenhouse-gas-emitting sector, accounting for roughly 38% of national emissions in 2024 — substantially higher than the EU average, reflecting the importance of dairy and beef farming to the Irish economy relative to other member states (EPA, 2025, *Greenhouse Gas Emissions and Projections*). Agriculture is also responsible for approximately 99% of Ireland's ammonia emissions, driven overwhelmingly by fertiliser use and manure management (CSO, 2025, *Environmental Accounts: Air Emissions*) — a striking illustration of how concentrated this particular externality is in a single sector of Irish business activity, and why it has become a central focus of Ireland's climate and land-use policy.

Positive Externality — Third-Level Education

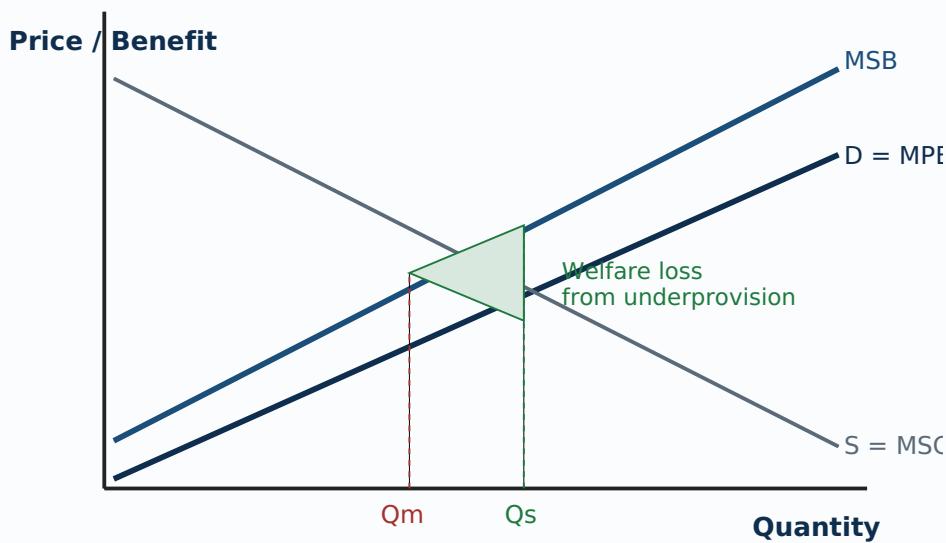


Figure 7: Positive consumption externality. The private market, guided only by MPB, settles at Q_m , underproducing relative to the social optimum Q_s (where $MSB = MSC$), since third parties' benefits are not reflected in individual students' willingness to pay.

The mechanism. A student enrolling in a third-level programme weighs their own Marginal Private Benefit (MPB) — higher future earnings, personal development, career opportunities. But the wider economy also benefits: a more educated workforce raises national productivity, supports innovation and entrepreneurship, and generates knowledge spillovers to employers, colleagues, and future generations that the individual student does not personally capture or get compensated for. Because the Marginal Social Benefit (MSB) of education exceeds its MPB, the market **underprovides** education relative to the socially optimal level, as Figure 7 shows: the private outcome (Q_m) falls short of the social optimum (Q_s), representing a welfare loss from underprovision.

Relevance to Irish business and the economy. A new postgraduate programme — a new Master's programme at UCC, for instance — generates exactly this kind of externality: the enrolled students benefit personally, but the wider Cork business community and Irish economy also benefit from a larger pool of skilled graduates, supporting the many multinational and indigenous firms that rely on a well-educated local labour force to locate and expand operations in the region.

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
17-20%	17-20%	Both a positive and a negative externality identified with genuine Irish business examples, each correctly diagrammed (MPC/MSB for the negative; MPB/MSB for the positive), the over/underproduction mechanism clearly explained for each, and specific supporting data or citations used.
12-16%	12-16%	Both externalities identified and explained with reasonable examples, but one diagram missing or unclear, or limited supporting data.
7-11%	7-11%	Externalities named but the private/social divergence mechanism not clearly explained; examples vague or not clearly Irish/business-specific.
3-6%	3-6%	Only one externality addressed, or significant conceptual confusion.
0-2%	0-2%	No meaningful attempt.

B2 (ii)**20%**

(ii) Explain two actions the Irish Government could take to manage the positive externality you identified in part (i) for the benefit of society.

FULL MODEL ANSWER

Since third-level education is underprovided by the free market ($Q_m < Q_s$ in Figure 7), the appropriate policy response is to **internalise the externality** — effectively lowering the private cost of education so that individual decisions move closer to the socially optimal level of provision.

1. Subsidies and Grants for Third-Level Education

Mechanism. By directly subsidising the cost of education — through free or reduced tuition fees, and means-tested maintenance grants such as those administered by SUSI (Student Universal Support Ireland) — the government lowers the effective private cost students face. This shifts the effective demand for education outward (or equivalently, reduces the price barrier), pushing actual enrolment from Q_m closer to the socially optimal Q_s , since more students for whom the private benefit alone might not have justified the cost will now find it worthwhile to enrol once the wider social benefit is reflected in a lower price.

Why this directly targets the externality. Because the underlying problem is that individual students don't capture (and aren't compensated for) the spillover benefits their education creates for the wider economy, a subsidy funded by general taxation effectively has the rest of society pay for part of the benefit it receives — correcting the gap between MPB and MSB directly at the point of decision.

2. Tax Relief and Investment in Third-Level Infrastructure and Places

Mechanism. The government can offer tax relief on tuition fees for postgraduate or part-time study, and can directly fund the expansion of third-level capacity — new lecture theatres, additional academic staff, and additional programme places (such as a new Master's programme at UCC) — ensuring the socially optimal quantity of education (Q_s) is actually available for students to take up, rather than being constrained by a shortage of supply even where demand exists.

Why supply-side investment matters alongside demand-side subsidy.

Subsidising the price of education only shifts effective demand toward Q_s if enough places actually exist to be filled at that price — without corresponding investment in capacity, a subsidy alone could simply create excess demand for a fixed number of places (as reflected in competitive points requirements for oversubscribed courses) rather than genuinely raising the quantity of education provided toward the social optimum.

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
17–20%	17–20%	Two genuinely distinct government actions explained (e.g. demand-side subsidy/grants and supply-side capacity investment), each explicitly linked back to closing the gap between private and social benefit identified in part (i), with specific Irish examples (SUSI, tax relief, capacity expansion).
12–16%	12–16%	Two actions explained but the connection back to the specific externality mechanism from part (i) is not made explicit.
7–11%	7–11%	One action well explained; the second vague or generic.
3–6%	3–6%	Actions named without explanation of mechanism.
0–2%	0–2%	No meaningful attempt.

B2 (iii)**20%**

(iii) Suggest, using examples, two ways in which ‘nudging’ could be used to manage the negative externality you outlined in part (i) for the benefit of society.

FULL MODEL ANSWER

Nudging works by redesigning the choice environment — through defaults, framing, or information — to steer behaviour in a socially beneficial direction, without restricting or banning any option or imposing a tax. Two nudges are particularly well suited to reducing agricultural ammonia and greenhouse gas emissions.

1. Framing and Social Comparison for Slurry Spreading and Fertiliser Practices

Mechanism. Ireland has already achieved meaningful reductions in ammonia emissions through increased uptake of low-emission slurry spreading (LESS) techniques and protected urea fertiliser — in 2023, greater use of low-emission slurry spreading alone avoided an estimated 7,000 tonnes of ammonia emissions (EPA, 2025). A nudge-based approach would frame this information competitively and socially: sending farmers a report comparing their own farm's emissions-reducing practices against the regional or county average (“80% of farms in your area have already switched to low-emission slurry spreading”) leverages social norms and herd mentality to accelerate voluntary adoption, without requiring any new regulation or subsidy.

Why this counts as a nudge rather than a mandate. Farmers retain complete freedom to continue their existing practices — the nudge simply makes the socially desirable choice more salient and socially rewarded, exploiting the same herd mentality and social comparison effects documented in behavioural economics more generally.

2. Defaults in Agricultural Advisory and Grant Scheme Enrolment

Mechanism. Where farmers interact with Department of Agriculture or Teagasc advisory schemes (for instance, renewing REPS/ACRES-style environmental schemes or nutrient management planning), enrolment in emissions-reducing options — protected urea use, reduced stocking density measures, or slurry application timing restrictions — could be set as the **default** option, with farmers free to opt out, rather than requiring farmers to actively opt in. Behavioural evidence consistently shows default options have a powerful effect on take-up simply because inertia favours whatever is pre-selected (Johnson & Goldstein, 2003, on default effects in organ donation registration, is a widely cited demonstration of the same underlying mechanism applied to a different policy context).

Why defaults are especially effective here. Given that many emissions-reducing practices require no fundamentally different farming approach (merely a different fertiliser type or spreading method), the barrier to adoption is often simply inertia rather than genuine cost or difficulty — exactly the situation where a default-option nudge tends to have the largest effect, since it removes the small but real friction of having to actively choose to opt in.

A complete answer explains *why* each nudge works (the specific behavioural mechanism — social comparison/herd mentality, or default/status-quo bias) rather than simply proposing an information campaign or policy change in general terms; nudging is specifically about how choices are presented and structured, not about changing the underlying incentives through taxes or regulation.

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
17-20%	17-20%	Two genuinely distinct nudges proposed (e.g. social comparison/framing, and a default-option redesign), each with the specific behavioural mechanism explained and applied concretely to the negative externality identified in part (i), with real-world grounding.
12-16%	12-16%	Two nudges proposed and reasonably explained, but the behavioural mechanism underlying one is not clearly articulated.
7-11%	7-11%	One genuine nudge well explained; the second is really a tax, subsidy, or regulation mislabelled as a nudge.
3-6%	3-6%	Nudging concept misunderstood or conflated with traditional regulation throughout.
0-2%	0-2%	No meaningful attempt.

Question B2 Total: 60%